

ESE5180: Lab 0 Zephyr

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GitHub Repository URL: https://github.com/lerntuspel/Lab_0-F26_Zephyr.git

1. Hello (Vanilla) Zephyr

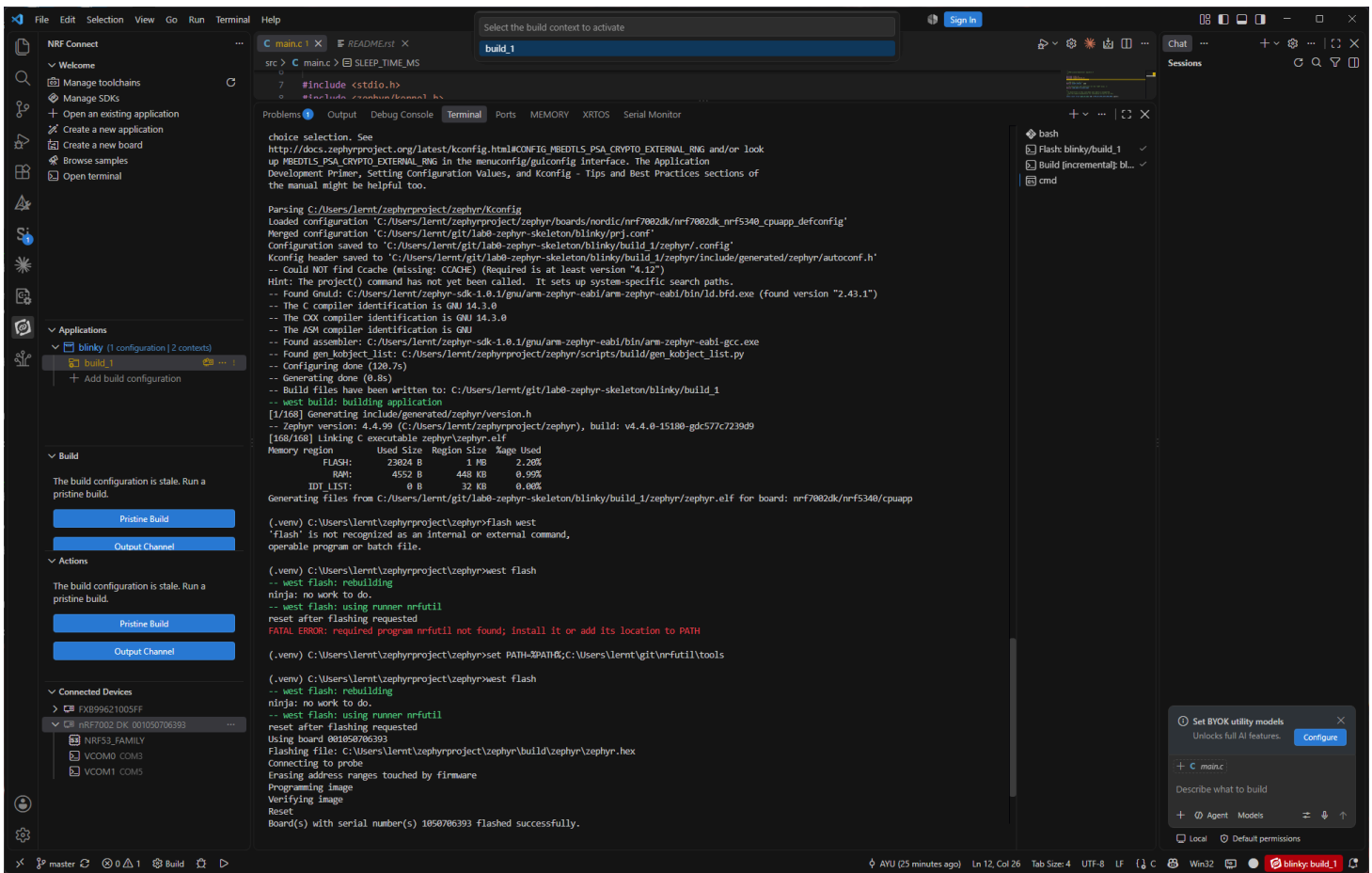
Done

2. Hello (Nordic) Zephyr

in commit hash ef7803644fee5784c8a5d0dab9a02888ec5e0b12

3. Building with West

3.1



4. Kconfig

I skimmed the reading

5. Device Tree (DT)

5.1

Also on commit hash ef7803644fee5784c8a5d0dab9a02888ec5e0b12

5.2 & 5.3

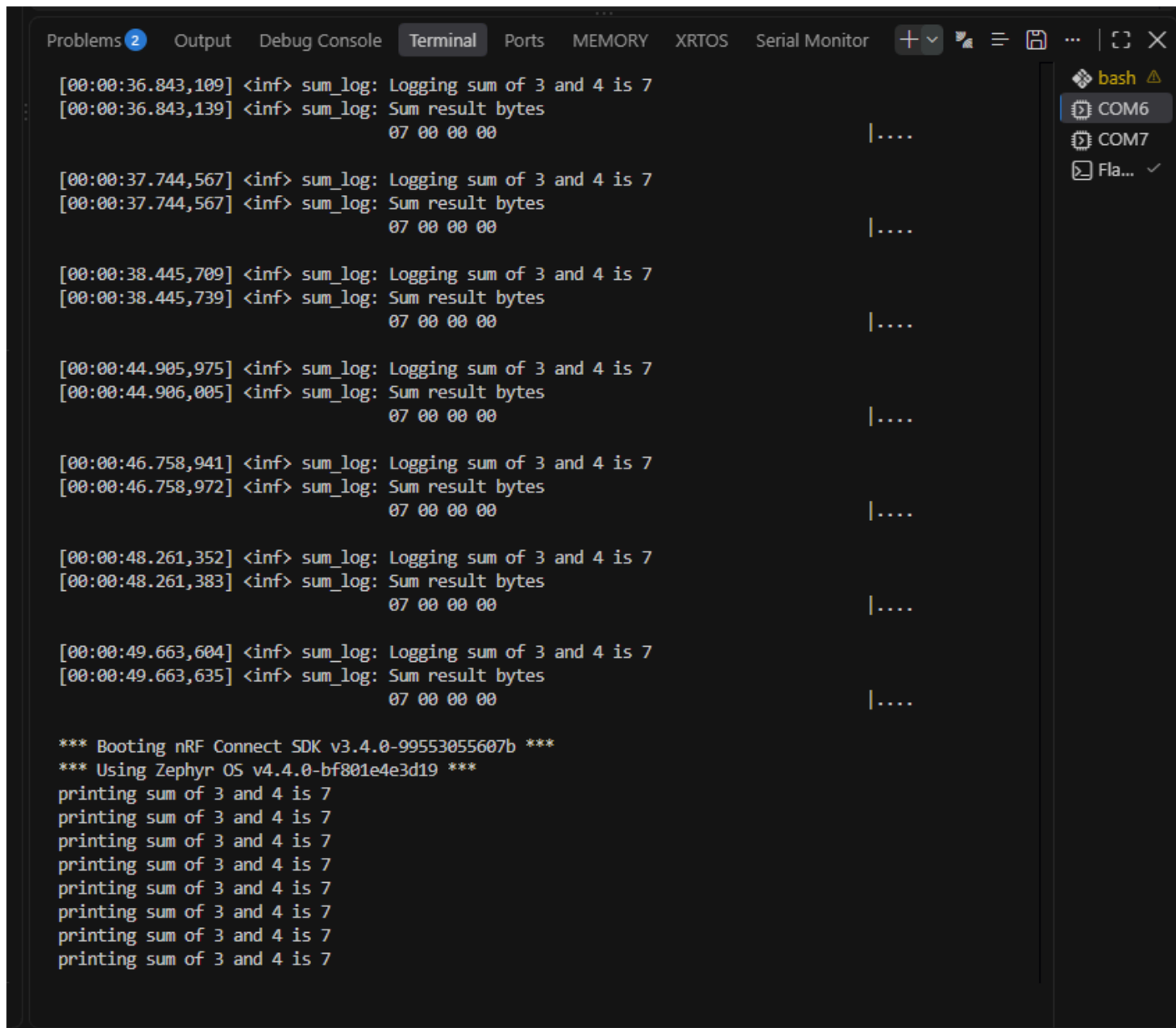
Commit hash 06138731ccd2648a9acc08f472f124b2646a6457

Why create an overlay if you can edit the dts?

The DTS is shared among all devices, the overlay maps the device tree to the specific device you are building for.

6. Printing vs. Logging

6.1



6.2

Done

6.3

Commit hash 17c0f905430f851943e7426fb4c99ab341412981

7. Ztest for Unit Testing

7.1

Commit hash 11a42670ef06a59590a6dd109853fdd0e120f338

7.2

Problems 3 Output Debug Console Terminal Ports MEMORY XRTOS Serial Monitor

----- TESTSUITE SUMMARY END -----

=====

PROJECT EXECUTION SUCCESSFUL
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE sum_log_test_suite
=====

START - test_sum_log_basic
I: Logging sum of 2 and 3 is 5
I: Sum result bytes
I: 05 00 00 00 |....

Assertion succeeded at CMAKE_SOURCE_DIR/src/main.c:36 (sum_log_test_suite_test_sum_log_basic)
PASS - test_sum_log_basic in 0.017 seconds
=====

START - test_sum_log_negative
I: Logging sum of -4 and -6 is -10
I: Sum result bytes
I: f6 ff ff ff |....

Assertion succeeded at CMAKE_SOURCE_DIR/src/main.c:46 (sum_log_test_suite_test_sum_log_negative)
PASS - test_sum_log_negative in 0.018 seconds
=====

START - test_sum_log_zero
I: Logging sum of 0 and 0 is 0
I: Sum result bytes
I: 00 00 00 00 |....

Assertion succeeded at CMAKE_SOURCE_DIR/src/main.c:56 (sum_log_test_suite_test_sum_log_zero)
PASS - test_sum_log_zero in 0.017 seconds
=====

TESTSUITE sum_log_test_suite succeeded

----- TESTSUITE SUMMARY START -----

SUITE PASS - 100.00% [sum_log_test_suite]: pass = 3, fail = 0, skip = 0, total = 3 duration = 0.052 seconds
- PASS - [sum_log_test_suite.test_sum_log_basic] duration = 0.017 seconds
- PASS - [sum_log_test_suite.test_sum_log_negative] duration = 0.018 seconds
- PASS - [sum_log_test_suite.test_sum_log_zero] duration = 0.017 seconds

----- TESTSUITE SUMMARY END -----

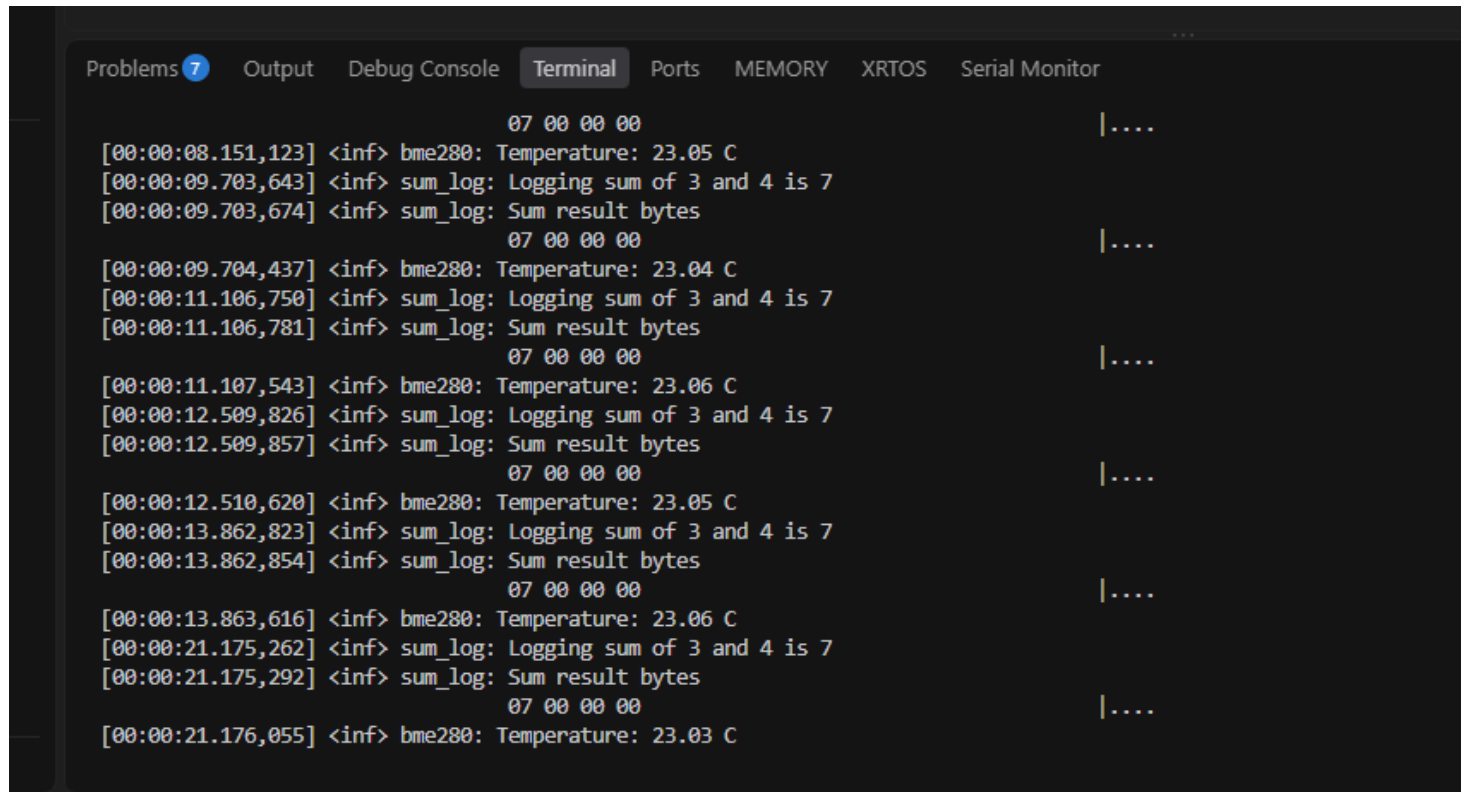
=====

PROJECT EXECUTION SUCCESSFUL
[]

CO...
FL...
bash
cmd

8. Adding a Peripheral (BME280)

8.1



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there is a navigation bar with tabs: Problems (7), Output, Debug Console, Terminal (selected), Ports, MEMORY, XRTOS, and Serial Monitor. The terminal displays a series of log messages. The messages are grouped into four sets, each containing three lines: a timestamped log entry for 'bme280: Temperature', a log entry for 'sum_log: Logging sum of 3 and 4 is 7', and a log entry for 'sum_log: Sum result bytes'. Each 'Sum result bytes' line is followed by the hex value '07 00 00 00' and a vertical bar followed by four dots '|....'.

```
Problems 7 Output Debug Console Terminal Ports MEMORY XRTOS Serial Monitor

                                07 00 00 00                                |....
[00:00:08.151,123] <inf> bme280: Temperature: 23.05 C
[00:00:09.703,643] <inf> sum_log: Logging sum of 3 and 4 is 7
[00:00:09.703,674] <inf> sum_log: Sum result bytes
                                07 00 00 00                                |....
[00:00:09.704,437] <inf> bme280: Temperature: 23.04 C
[00:00:11.106,750] <inf> sum_log: Logging sum of 3 and 4 is 7
[00:00:11.106,781] <inf> sum_log: Sum result bytes
                                07 00 00 00                                |....
[00:00:11.107,543] <inf> bme280: Temperature: 23.06 C
[00:00:12.509,826] <inf> sum_log: Logging sum of 3 and 4 is 7
[00:00:12.509,857] <inf> sum_log: Sum result bytes
                                07 00 00 00                                |....
[00:00:12.510,620] <inf> bme280: Temperature: 23.05 C
[00:00:13.862,823] <inf> sum_log: Logging sum of 3 and 4 is 7
[00:00:13.862,854] <inf> sum_log: Sum result bytes
                                07 00 00 00                                |....
[00:00:13.863,616] <inf> bme280: Temperature: 23.06 C
[00:00:21.175,262] <inf> sum_log: Logging sum of 3 and 4 is 7
[00:00:21.175,292] <inf> sum_log: Sum result bytes
                                07 00 00 00                                |....
[00:00:21.176,055] <inf> bme280: Temperature: 23.03 C
```

8.2

```
[00:00:09.972,595] <inf> sum_log: Sum result bytes
                                07 00 00 00
[00:00:09.973,358] <inf> bme280: Temperature: 26.03 C
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE devicetree_sanity
=====
START - test_bme280_address
PASS - test_bme280_address in 0.001 seconds
=====
START - test_bme280_node_okay
PASS - test_bme280_node_okay in 0.001 seconds
=====
START - test_button_alias_okay
PASS - test_button_alias_okay in 0.001 seconds
=====
START - test_button_gpio_ready
PASS - test_button_gpio_ready in 0.001 seconds
=====
START - test_i2c1_bus_okay
PASS - test_i2c1_bus_okay in 0.001 seconds
=====
START - test_i2c_bus_ready
PASS - test_i2c_bus_ready in 0.001 seconds
=====
START - test_led_alias_okay
PASS - test_led_alias_okay in 0.001 seconds
=====
START - test_led_gpio_ready
PASS - test_led_gpio_ready in 0.001 seconds
=====
TESTSUITE devicetree_sanity succeeded

----- TESTSUITE SUMMARY START -----

SUITE PASS - 100.00% [devicetree_sanity]: pass = 8, fail = 0, skip = 0, total = 8 duration = 0.008 seconds
- PASS - [devicetree_sanity.test_bme280_address] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_bme280_node_okay] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_button_alias_okay] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_button_gpio_ready] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_i2c1_bus_okay] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_i2c_bus_ready] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_led_alias_okay] duration = 0.001 seconds
- PASS - [devicetree_sanity.test_led_gpio_ready] duration = 0.001 seconds

----- TESTSUITE SUMMARY END -----

=====
PROJECT EXECUTION SUCCESSFUL
```